

SharmaDSC520Week8.2

2025-01-30

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats   1.0.0     v stringr   1.5.1
## v lubridate 1.9.4     v tibble   3.2.1
## v purrr     1.0.2     v tidyr    1.3.1
## v readr     2.1.5
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

##   Package LibPath Version Priority Depends Imports LinkingTo Suggests
##   Enhances License License_is_FOSS License_restricts_use OS_type Archs
##   MD5sum NeedsCompilation Built

## Read Housing CSV file for houses
df_housing = read.csv("week-6-housing.csv")

#any missing values in housing data
anyNA(df_housing)

## [1] FALSE

colSums(is.na(df_housing))

##           Sale.Date           Sale.Price           sale_reason
##                0                0                0
##   sale_instrument   sale_warning   sitetype
##                0                0                0
##       addr_full           zip5       ctyname
##                0                0                0
##       postalctyn           lon           lat
##                0                0                0
##   building_grade square_feet_total_living bedrooms
##                0                0                0
##   bath_full_count   bath_half_count   bath_3qtr_count
##                0                0                0
##       year_built       year_renovated   current_zoning
##                0                0                0
```

```

##          sq_ft_lot          prop_type          present_use
##              0              0              0
# Data cleaning

df_housing <- df_housing %>%

  mutate(across(c(bedrooms,bath_full_count,bath_half_count,bath_3qtr_count,ctyname), ~replace(., is.na(

#Impute missing values
  filter(Sale.Price < quantile(Sale.Price, 0.95)) %>% # Remove extreme outliers

#Impute missing values
  filter(present_use < 29) # Remove extreme outliers

## Warning: There was 1 warning in `mutate()`.
## i In argument: `across(...)`
## Caused by warning in `mean.default()`:
## ! argument is not numeric or logical: returning NA

# create linear regression model where "sq_ft_lot" predicts Sale Price
modell1 <- lm(Sale.Price ~ sq_ft_lot+bedrooms+year_built, df_housing)

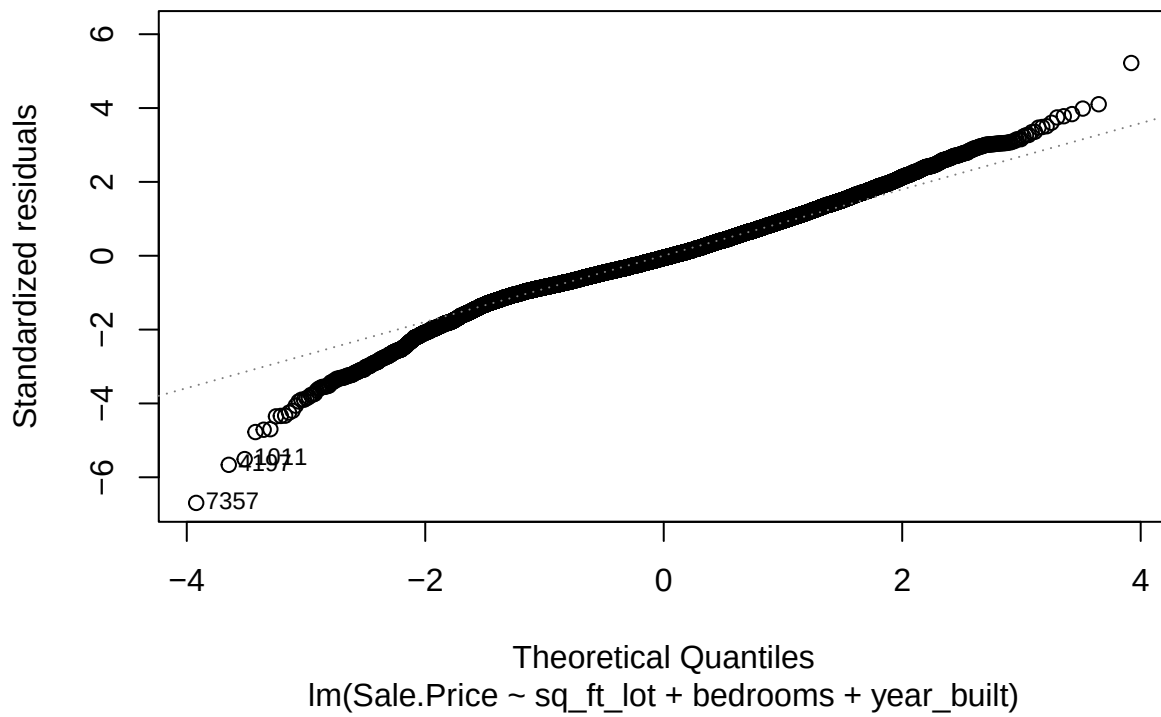
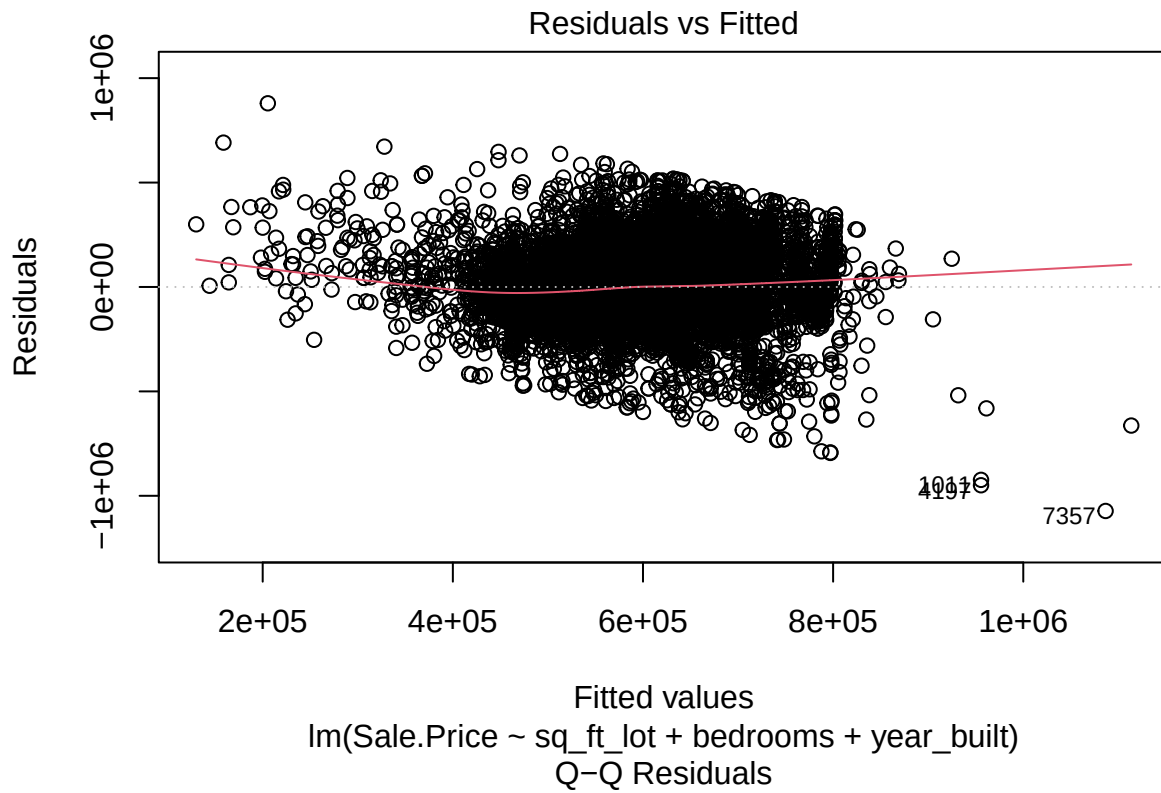
# This model is based on the total sq ft lot with bedrooms and year built of the house including its lot

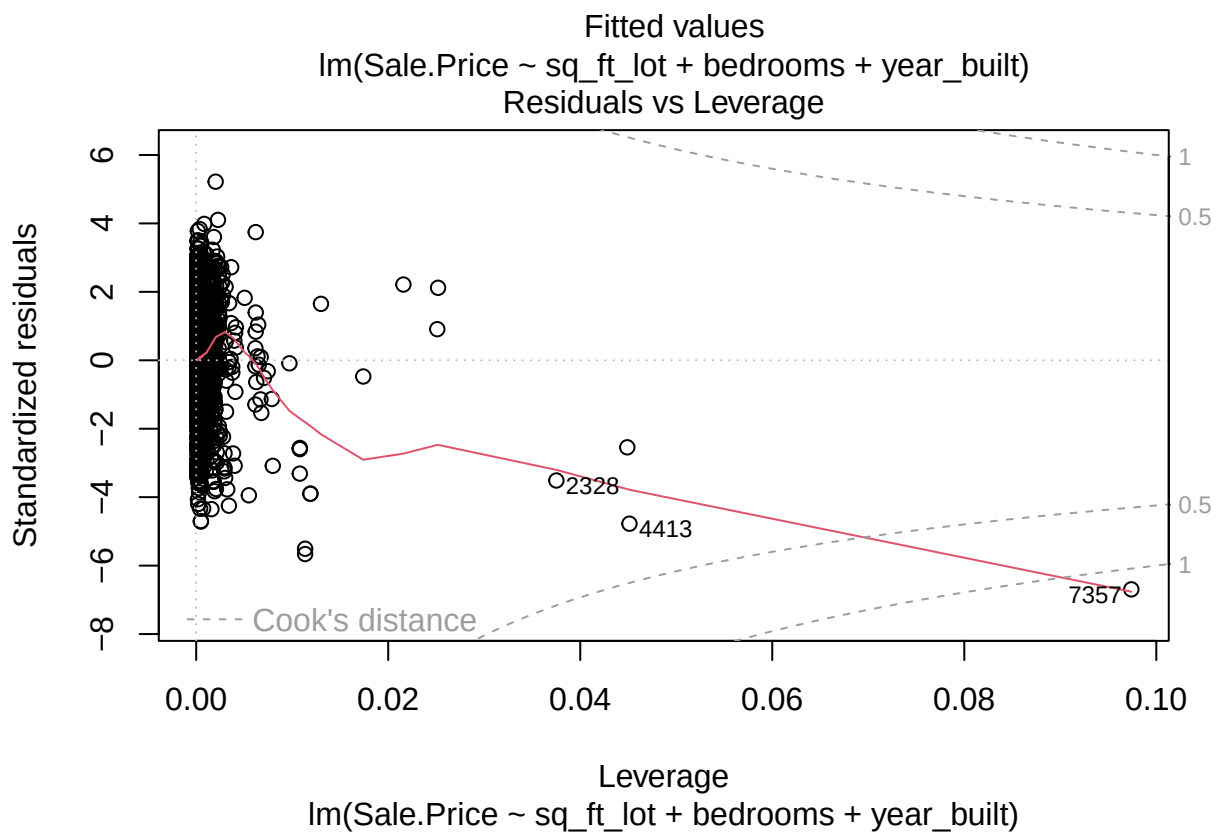
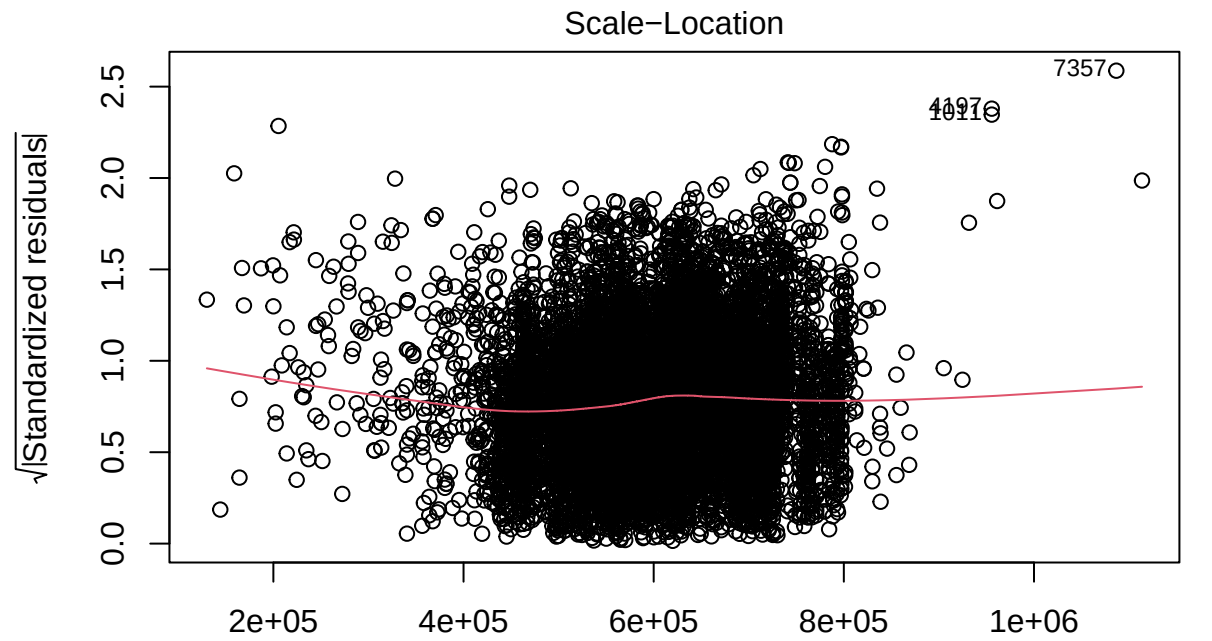
# Summarize the model
summary(modell1)

##
## Call:
## lm(formula = Sale.Price ~ sq_ft_lot + bedrooms + year_built,
##     data = df_housing)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1072584  -101360    -9625   102663   879586
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.315e+06  1.854e+05 -44.861  < 2e-16 ***
## sq_ft_lot    2.263e-01  3.261e-02   6.941  4.1e-12 ***
## bedrooms     6.714e+04  1.913e+03  35.099  < 2e-16 ***
## year_built   4.357e+03  9.313e+01  46.786  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 168700 on 11339 degrees of freedom
## Multiple R-squared:  0.242, Adjusted R-squared:  0.2418
## F-statistic: 1207 on 3 and 11339 DF, p-value: < 2.2e-16

# Plot the model
plot(modell1)

```





Interpretation:

The coefficient for "sq_ft_lot" represents the average increase in Sale Price for every additional sq

The R-squared value indicates the proportion of variance in Sale Price explained by the "sq_ft_lot" v

Explanation of key metrics:

R-squared (0.242):

This indicates that the model explains 24.2% of the variance in housing prices based on the included variables, suggesting a relatively good fit.

Adjusted R-squared (0.2418):

Similar to R-squared, but penalizes for the number of variables in the model, providing a more accurate measure when comparing models with different predictor counts.

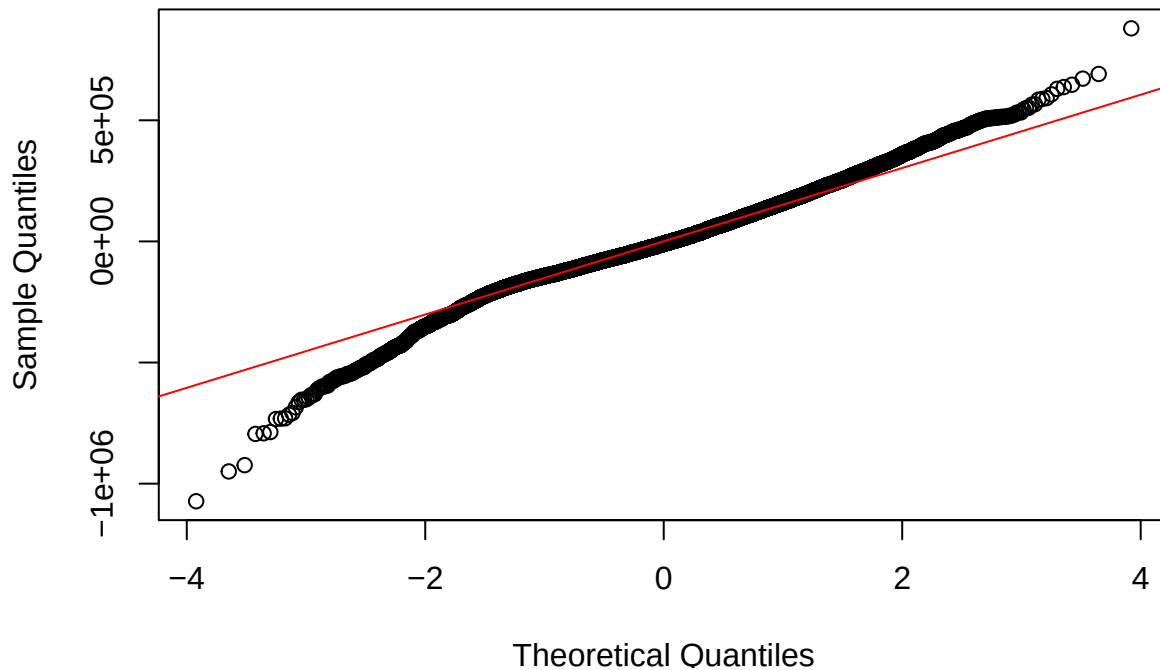
```
# Assuming you have a linear regression model called "housing_model"
residuals1 <- resid(model1)

# Plot residuals against predicted values
plot(model1$fitted.values, residuals1,
     xlab = "Predicted Housing Price", ylab = "Residuals",
     main = "Residual Plot")
```



```
# Create a QQ plot
qqnorm(residuals1)
qqline(residuals1,col="red")
```

Normal Q-Q Plot



Interpretation:

If the points closely follow the red line, the residuals are approximately normal [1,3,6].

If the points deviate significantly from the line, especially at the tails, normality is not met [1,2,4].

```
# Create a linear regression model predicting Sale Price using multiple variables
model2 <- lm(Sale.Price ~
  bedrooms
  + bath_full_count
  + bath_half_count
  + bath_3qtr_count
  + sq_ft_lot
  + year_built,
  df_housing)

# Summary of the model
summary(model2)
```

```
##
## Call:
## lm(formula = Sale.Price ~ bedrooms + bath_full_count + bath_half_count +
##     bath_3qtr_count + sq_ft_lot + year_built, data = df_housing)
##
## Residuals:
```

##	Min	1Q	Median	3Q	Max
##	-2042433	-95272	-6722	97913	869521

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -5.714e+06  2.118e+05 -26.978  < 2e-16 ***
## bedrooms      3.913e+04  2.184e+03  17.912  < 2e-16 ***
## bath_full_count 7.603e+04  3.370e+03  22.563  < 2e-16 ***
## bath_half_count 5.457e+04  3.476e+03  15.701  < 2e-16 ***
## bath_3qtr_count 5.744e+04  3.212e+03  17.881  < 2e-16 ***
## sq_ft_lot      1.403e-01  3.189e-02   4.399  1.1e-05 ***
## year_built     3.003e+03  1.074e+02  27.961  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 163900 on 11336 degrees of freedom
## Multiple R-squared:  0.2845, Adjusted R-squared:  0.2841
## F-statistic: 751.2 on 6 and 11336 DF,  p-value: < 2.2e-16

# Example prediction for a new house
new_house <- data.frame(
  bedrooms = 3,
  bath_full_count = 2,
  bath_half_count = 1,
  bath_3qtr_count = 0,
  sq_ft_lot = 3840,
  year_built = 2012
)

predicted_price <- predict(model2, newdata = new_house)
print(predicted_price)

##          1
## 652567.3
```

Explanation:

- “SalePrice ~”: specifies Sale Price as the dependent variable

bedrooms + bath_full_count + bath_half_count + bath_3qtr_count + sq_ft_lot and the new variable year_renovated meaning the model differences based on

year_built vs other variables

-These are the predictor variables,including:

bedrooms,bath_full_count,bath_half_count,bath_3qtr_count,sq_ft_lot

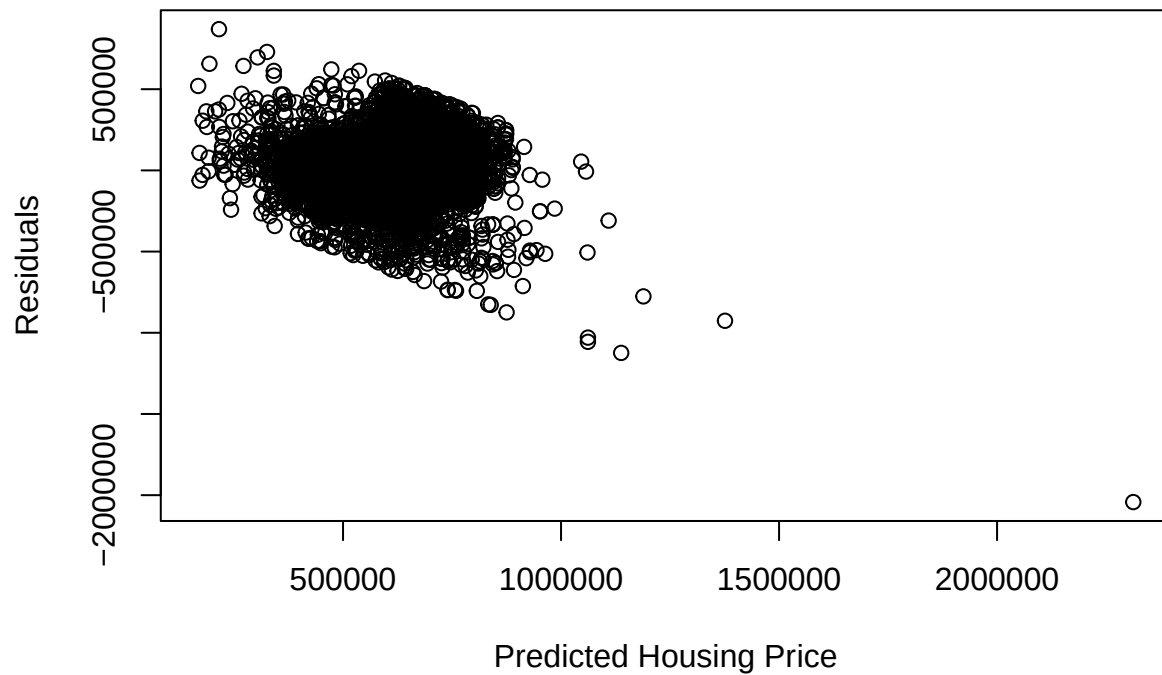
those give prediction related to how much will be the value of the house.

```
# Assuming you have a linear regression model called "housing_model"
residuals2 <- resid(model2)

# Plot residuals against predicted values
plot(model2$fitted.values,residuals2,
      xlab = "Predicted Housing Price", ylab = "Residuals",
```

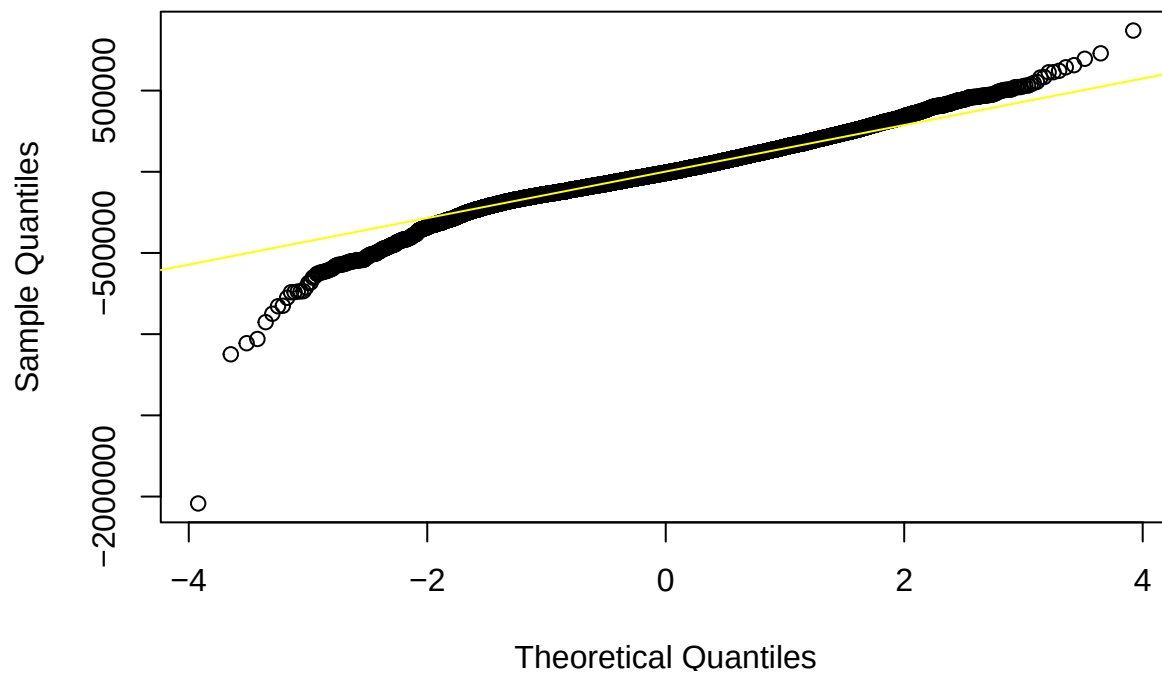
```
main = "Residual Plot")
```

Residual Plot



```
# Create a QQ plot  
qqnorm(residuals2)  
qqline(residuals2,col="yellow")
```

Normal Q-Q Plot



Explanation of key metrics:

R-squared (0.2845):

This indicates that the model explains 28.5% of the variance in housing prices based on the included variables, suggesting a relatively good fit.

Adjusted R-squared (0.2841):

A higher R-squared in the second model indicates it explains a larger proportion of the variance in house prices.

```
anova(model1,model2)

## Analysis of Variance Table
##
## Model 1: Sale.Price ~ sq_ft_lot + bedrooms + year_built
## Model 2: Sale.Price ~ bedrooms + bath_full_count + bath_half_count + bath_3qtr_count +
##          sq_ft_lot + year_built
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1   11339 3.2257e+14
## 2   11336 3.0448e+14   3 1.8086e+13 224.46 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

These models are biased model1 or model2 or both as these are having deviation from normal distribution in the residual shows potential bias in the model.

```
preds1 <- predict(model1, df_housing)

preds2 <- predict(model2, df_housing)

# Calculate RMSE
rmse1 <- rmse(df_housing$Sale.Price, preds1)
print(paste("RMSE:", rmse1))

## [1] "RMSE: 168634.108233226"

rmse2 <- rmse(df_housing$Sale.Price, preds2)
print(paste("RMSE:", rmse2))

## [1] "RMSE: 163838.247870101"
```